Human GLP1R designs: the final 20

Final computational report | 25 October 2026

This report presents 20 protein designs selected for experimental evaluation against human GLP1R. Six are original LigandForge designs. Fourteen combine two previously designed binding domains with a new linker generated by AP Novo. The report gives every complete sequence, its predicted properties, how it was created, and the inputs and programs used to evaluate it.

All 20 passed two computational thresholds: an interface-confidence score, ipSAE, above 0.895 and a DeltaForge predicted binding affinity, Kd, below 10 nM. Candidates were then ranked by ipSAE, with the highest 20 retained. These predictions help prioritize experiments. Binding and receptor activation have not yet been demonstrated experimentally.

1 Design strategy and selection

GLP1R is the human glucagon-like peptide-1 receptor. The goal is to identify proteins that bind the receptor and could be tested for receptor activation. The fusion designs connect a domain aimed at the receptor's extracellular domain (ECD) to an original LigandForge design. This creates a hypothesis of coordinated engagement at two receptor regions; the calculations do not demonstrate that both domains engage as intended or that the receptor is activated.

The original designs were predicted against a 409-residue receptor construct. The fusion designs used a 440-residue construct. These different inputs matter when comparing the two groups. Their shared ranking follows the chosen computational selection rule; it is not a controlled experimental comparison.

1.1 Designs evaluated

Stage	Completed work	How it contributed to the final selection
Original LigandForge designs	1,000 sequences and 1,000 Boltz2 Exact complex predictions	Supplied original candidates and one parent domain for fusions
Active fusion campaign	183 parent pairs, ten linker attempts each: 1,830 generated fusions and 1,830 Exact predictions	Supplied fusion candidates; reused pilot predictions were counted once
Extracellular-position screen	1,725 fusion models retained; 105 set aside	Removed models whose free C terminus failed the extracellular-position check
Affinity evaluation	50 original designs and 50 fusions	Supplied the two groups from which the final 20 were selected
Final selection	25 passed both thresholds; the highest 20 by ipSAE were retained	Six originals and fourteen fusions, all with different full sequences

Two prohibited parent pairs, P000 and P001, were excluded from the active fusion campaign because their selected free C termini were on the intracellular side. The 1,830 active predictions include 20 reused pilot predictions and 1,810 later predictions. Other design and receptor-sampling work occurred during the project, but the counts above describe the branches contributing to this selection, rather than every project attempt.

1.2 Ranking rule

The strict thresholds were ipSAE greater than 0.895 and predicted Kd less than 10 nM. Among qualifying designs, higher ipSAE determined final rank. The predefined tie-breakers were lower predicted Kd, then design group and design name; no tie affected the cutoff. There was no diversity-based reordering or fixed quota for either group. A candidate's earlier rank within its own group is kept separate from its final rank.

2 Programs, inputs and evaluation methods

2.1 Receptor sequences and target regions

The target was human GLP1R, UniProt P43220-1. The challenge reference structure was PDB 6X18, author chain R. Original-design complex predictions used wild-type receptor residues 24-432, totaling 409 amino acids. ECD-parent and fusion predictions used residues 24-463, totaling 440 amino acids. Both complete receptor sequences are printed in section 7. In the final complex models, the candidate is chain A and the receptor is chain B. Receptor position B1 corresponds to UniProt position 24; add 23 to convert any native receptor position to its UniProt number.

The original LigandForge request specified 27 receptor-pocket positions: 141, 144, 145, 148, 152, 190, 194, 197, 201, 230, 233, 234, 237, 241, 298, 299, 300, 306, 309, 310, 313, 372, 380, 384, 387, 388 and 391, in the submitted structure's numbering. Pocket expansion was set to zero. The provider's input-preparation software instead reported fourteen positions, 60, 61, 76-80 and 105-111, in a different numbering system. Its receptor structure differed from the submitted structure. The available records do not establish the correspondence between these lists, so the requested pocket cannot be assumed to describe the actual generation input precisely.

The ECD-parent route used an open-receptor computational model identified as seed 156, sample 1. It targeted Arg40, which is receptor position B17. Gln37 (B14) and Glu34 (B11) were additional monitored positions. Only the Arg40 contact was required in the downstream contact screen; the other two were optional.

2.2 LigandForge: original sequence generation

LigandForge version 6.5.21 generated the original population. Its input-preparation software was version 6.5.13. The saved configuration requested a binding segment of 20-60 amino acids, followed by the fixed six-residue sequence GSGSGS and four terminal residues. Complete exported designs were 32-70 amino acids long. A separate provider summary describes a 20-40-residue binding segment; the reason for this discrepancy is unknown. Each selected sequence is printed in full, so its actual composition is unambiguous.

Generation used cell-free-expression guidance and recombinant-only residues. Quality-guided sampling and extracellular-domain trimming were disabled. No charge restriction, paralog target or pH condition was specified. The selected original designs retain their complete exported sequences, including the GSGSGS region and terminal residues. No mutation, shortening or additional cyclization chemistry was introduced during report preparation.

Original generated complex structures were not retained. Consequently, the later Boltz2 Exact structures cannot be compared with an original LigandForge structural reference. The provider also exported creation timestamps later than a recorded folding-completion timestamp. Their literal order is inconsistent and is not used to infer the sequence of execution.

2.3 AP Novo: extracellular-domain parent generation

All ECD parents represented in the final 20 were generated by AP Novo. The input was the complete 440-residue open-receptor model, with no starting binder, GLP1 peptide, explicit membrane or glycan atoms. Generation used 1,000 sampling steps. Selected parent conditions requested new binders of either 20-40 or 40-60 amino acids.

The receptor sequence was fixed throughout. Some conditions also fixed all receptor coordinates. Other conditions allowed receptor positions B106-116 to move, while keeping their sequence fixed. This flexible receptor segment is different from the new linker later placed between the two binder domains. Before generation, CPU checks confirmed which atoms could move and which sequences had to remain fixed. Each candidate's history states its parent condition.

Boltz2 Fast provided an initial screen. The top 100 designs by Fast-ipSAE from each of twelve conditions were taken forward for standalone Boltz2 Exact prediction. Parent ranks and scores in the candidate histories refer to these standalone predictions. A high-scoring fusion can contain a parent that scored poorly alone; the fusion result does not validate that parent's standalone pose.

2.4 AP Novo: joining the two parent domains

Each fusion retains an ECD parent and a LigandForge parent intact, connected in the stated order. The two parent structures were positioned using their receptor context. Pair selection preferred an exposed free C terminus on the extracellular side, then the shorter gap between the first domain's C terminus and the second domain's N terminus. The exposure criterion required a terminal exposure score of at least 0.25 and an extracellular height above 19.7 Å in the oriented membrane reference.

The minimum requested linker length was the larger of eight residues or the junction gap divided by 2.8 Å, rounded upward, minus one residue. The maximum was twelve residues longer. Ten explicit random seeds were used per active pair. AP Novo generated the linker while holding both parent sequences and all retained parent atoms fixed. Each candidate history gives the actual linker sequence and length, allowed length range, orientation, seed and attempt number.

To fit the generator's input limit, the receptor context contained the complete ECD, native receptor positions 1-105, plus nearby receptor fragments. All contacts within 4.5 Å were mandatory. Additional context was selected within 10 Å with a one-residue flank; that radius could be reduced to keep the total input within 512 residues. Each candidate's input record below lists the exact retained receptor intervals and actual radius. Their sequences can be read directly from the 440-residue receptor sequence in section 7. Full 440-residue receptor context was restored for the subsequent Exact prediction.

2.5 Boltz2 Exact: predicting candidate-receptor complexes

Every reported final model was produced by Boltz2 Exact using the complete candidate sequence and the appropriate receptor sequence. Settings were random seed 101, three recycling steps, 200 sampling steps and one diffusion sample. Up to 8,192 sequences were allowed in the multiple-sequence alignment (MSA). The binder had no alignment beyond its own sequence. No structural templates or online MSA server were used. Target alignments and native input features were generated or validated on CPU before GPU prediction. The original target-alignment preparation used ColabFold 1.5.2 with UniRef30 version 2202 and no environmental-sequence search.

The principal ranking score, ipSAE, was calculated with the DunbrackLab implementation using a predicted alignment error cutoff of 10 Å and a distance cutoff of 15 Å. The larger of the two directional chain scores was used for selection. The individual directional scores remain visible for fusion designs. Interface confidence does not measure binding affinity or receptor activation.

The report also includes ipTM and, where available, pDockQ, pDockQ2 and the local interaction score (LIS). These summarize different aspects of the predicted interface and should not be treated as interchangeable probabilities. Local structural confidence, pLDDT, is given for the full fusion and its separate domains and linker. Fusion segment values use a 0-100 scale; the original population's saved complex mean uses a 0-1 scale and is labeled accordingly.

2.6 Structural consistency and extracellular-position checks

DockQ and TM-score compared fusion Exact structures with their computational design references. DockQ summarizes agreement in contacts and binding pose. TM-score summarizes structural similarity. Neither comparison uses an experimentally determined structure. A low value can therefore reveal that the predicted fusion differs substantially from its intended design even when its interface-confidence score is high. Root-mean-square deviation (RMSD) measures positional differences in Å; the receptor-core RMSD uses a fit to the receptor core, rather than an independent fit to the binder.

The membrane-position screen used the orientation of 6X18 in the Orientations of Proteins in Membranes (OPM) reference. A membrane half-thickness of 14.9 Å, a 3.0 Å outward margin and a 1.8 Å atomic-radius allowance gave the 19.7 Å extracellular threshold. Alignment used 253 receptor-core C-alpha atoms from native positions 117-234, 240-315, 338-346 and 351-400. The final selected fusions passed the post-prediction free-C-terminal height check. This does not establish that an attached assay tag would be accessible.

Additional screening settings included a 2.0 Å minimum binder-receptor heavy-atom distance, a 4.5 Å required Arg40 side-chain contact, and 15 Å exclusion zones around the ND2 atoms of potential glycosylation sites at UniProt positions 63, 82 and 115. These are geometric screens. Glycan occupancy was not established, and explicit glycans and membrane atoms were not simulated.

2.7 DeltaForge: predicting binding affinity

DeltaForge evaluated each complete candidate as one chain against one receptor chain from the saved Boltz2 Exact model. The intact fusion included both parent domains and the full linker. The original candidates retained the 409-residue receptor; fusions retained the 440-residue receptor. Structures were supplied in PDB format, and sequence and model correspondence were checked before joining the affinity results to the confidence scores.

All selected results used the DeltaForge v10.2 unified parallel scorer, dated 8 May 2026, with the Boltz2-calibrated gradient-boosted affinity model dated 29 March 2026. Kd is reported in nanomolar (nM); lower values indicate stronger predicted binding. No candidate-specific uncertainty interval was returned. Six selected fusions exceed 140 residues, placing them in the provider's longer-sequence category; this label does not establish how well the affinity model is calibrated for them.

The requests did not include the folding-confidence values needed for the provider's combined binder/non-binder classification, so that classification was left unassigned. The separately calculated ipSAE values do not change the saved provider response. The returned energy field had no explicit unit and was not used to recalculate Kd. Contact, hydrogen-bond, salt-bridge and hydrophobic-contact counts in the candidate tables are provider-defined geometric features, rather than experimentally verified interactions.

2.8 Reading the candidate records

Each candidate record identifies its complete sequence, parents when applicable, generation inputs, prediction settings and results. Parent names, random seeds, residue intervals and program versions are retained because they explain how the design was produced. Sequence and structural-input checks confirmed that each selected affinity result belonged to the intended complete molecule. This revision uses saved calculations; no new design, folding or affinity predictions were run.

3 Molecular views

Each candidate has a dedicated structure page after its metric table. The complete receptor-complex view is shown beside an interface detail. LigandForge domains are orange, AP Novo linkers purple, ECD-binding domains blue, receptor ECD teal, and the remaining receptor gray. Black sticks identify the annotated receptor target positions. The receptor ECD display covers native positions B1-105, corresponding to UniProt residues 24-128.

The full and detailed views use different zoom levels. The detailed view omits distant receptor regions for clarity; all residues remain in the model used for scoring. Predicted domain positions may differ from the intended two-site design. A favorable overall interface score alone cannot establish the intended engagement geometry or an active receptor state.

4 The selected 20

Rank	Design	Design route	Length	Receptor length	Rank within group	Exact ipSAE	Predicted Kd nM
1	LK_P050_07	Fusion	151	440	1	0.919	3.04
2	LK_P063_07	Fusion	140	440	3	0.911	1.34
3	LK_P109_06	Fusion	113	440	4	0.911	1.33
4	Design_520	Original	53	409	4	0.911	3.45
5	LK_P088_08	Fusion	153	440	5	0.908	0.718
6	LK_P050_03	Fusion	149	440	6	0.906	3.28
7	Design_973	Original	65	409	8	0.904	0.619
8	LK_P157_08	Fusion	139	440	7	0.902	3.24
9	LK_P069_04	Fusion	154	440	9	0.902	1.25
10	LK_P024_03	Fusion	128	440	10	0.902	3.79
11	Design_994	Original	70	409	11	0.901	2.74
12	LK_P012_08	Fusion	125	440	11	0.901	1.45
13	LK_P062_00	Fusion	146	440	12	0.900	1.13
14	Design_440	Original	54	409	13	0.900	4.4
15	LK_P141_07	Fusion	121	440	14	0.900	1.63
16	LK_P002_05	Fusion	104	440	16	0.899	0.813
17	LK_P012_04	Fusion	130	440	17	0.899	2.15
18	Design_787	Original	49	409	14	0.899	5.38
19	Design_15	Original	50	409	15	0.899	4.29
20	LK_P093_03	Fusion	159	440	19	0.898	0.823

All twenty sequences are unique. Lengths range from 49 to 159 aa. Six fusion candidates exceed 140 amino acids. The compact table rounds Kd for readability; each candidate section reports its prediction to six decimal places.

5.01 LK_P050_07

Final rank 1 | Two-domain fusion | 151 amino acids

Design history. AP Novo generated the ECD parent S11_02_089 under the condition requesting 40-60-residue binders with receptor positions 106-116 allowed to move. Its standalone Exact rank was 2, with ipSAE 0.927. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P050 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 608 for attempt 7. The allowed linker range was 25-37 residues; the selected linker contains 29 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-53; Linker: residues 54-82; LigandForge domain: residues 83-151. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
AFSPTSSEVAPDLPFVVLPYPPPPSSSVI
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 165-222, 253-262, 264-266, 268-296, 337-372, totaling 266 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 425 residues. The gap between the two domains before linker generation was 70.9 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 2, output 89. That batch began with random seed 301; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MGISALEEFDRLRRASILLSLIRSGSTGSGFDAFLEKSGVKGAPAFLAVPSSNAFSPTSSEVAPDLPF
VVLPYPPPPSSSVIEVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYAS
RYFSIGSGSGSQQFS
```

5.01 LK_P050_07 - predicted properties

Boltz2 Exact predicted the full 151-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 3.04 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.919
Boltz2 Exact	Interface confidence, ipTM	0.936
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.919 / 0.822
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	49.8 / 40.4
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	66.6 / 40.8
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.381 / 0.261 / 0.521
Design compared with Exact	Fusion DockQ	0.19
Design compared with Exact	Fusion TM-score	0.287
Design compared with Exact	Receptor-core RMSD (Å)	0.8
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	2.2 / 2.9 / 2.4
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	42.4
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	3.04
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	61 / 14 / 10 / 18
Within this design group	Predicted-affinity rank among the scored top 50	27

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.01 LK_P050_07 - bound structures

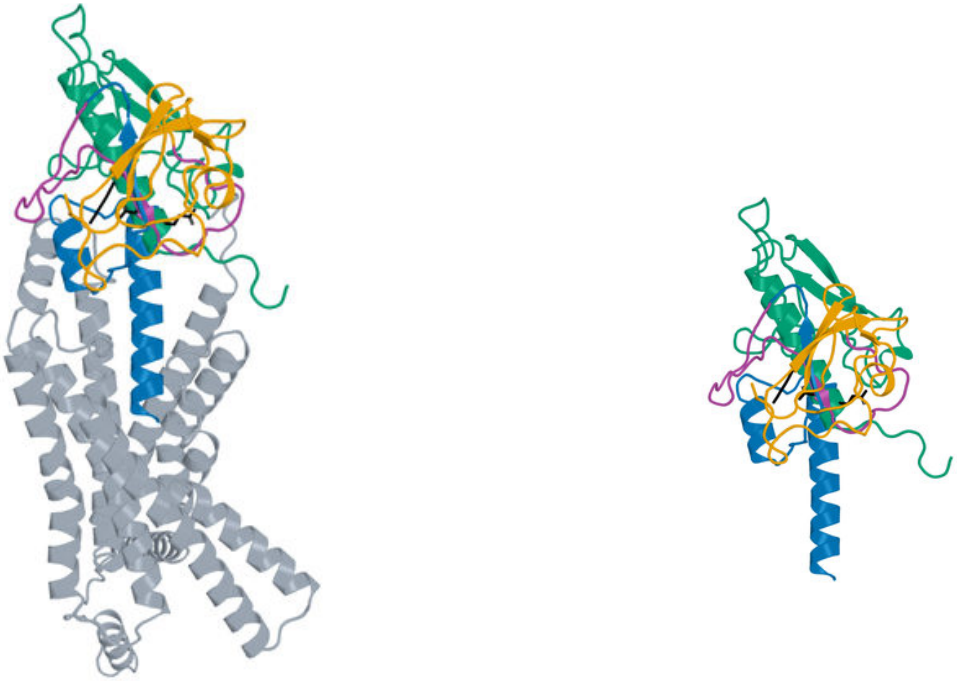


Figure 1. LK_P050_07, final rank 1. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.02 LK_P063_07

Final rank 2 | Two-domain fusion | 140 amino acids

Design history. AP Novo generated the ECD parent S08_03_083 under the condition requesting 20-40-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 15, with ipSAE 0.910. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P063 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 738 for attempt 7. The allowed linker range was 23-35 residues; the selected linker contains 31 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-40; Linker: residues 41-71; LigandForge domain: residues 72-140. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GMFVDPSEVLDPSPPPPPPLSLPPVPATDVV
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 166-219, 260-262, 264-292, 343-370, totaling 244 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 388 residues. The gap between the two domains before linker generation was 65.9 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 3, output 83. That batch began with random seed 401; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MEKKREDFKEVLKFRYSRYPSLDRFVLVNTAEESIRHLLIGMFVDPSEVLDPSPPPPPPLSLPPVPAT
DVVEVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYASRYFSIGSGSGS
QQFS
```

5.02 LK_P063_07 - predicted properties

Boltz2 Exact predicted the full 140-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.34 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.911
Boltz2 Exact	Interface confidence, ipTM	0.889
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.911 / 0.660
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	48.6 / 43.9
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	67.0 / 40.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.372 / 0.214 / 0.541
Design compared with Exact	Fusion DockQ	0.211
Design compared with Exact	Fusion TM-score	0.356
Design compared with Exact	Receptor-core RMSD (Å)	1.1
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.8 / 5.2 / 3.0
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	60.5
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.4 / 0
DeltaForge	Predicted Kd (nM)	1.34
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	109 / 8 / 6 / 31
Within this design group	Predicted-affinity rank among the scored top 50	13

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.02 LK_P063_07 - bound structures

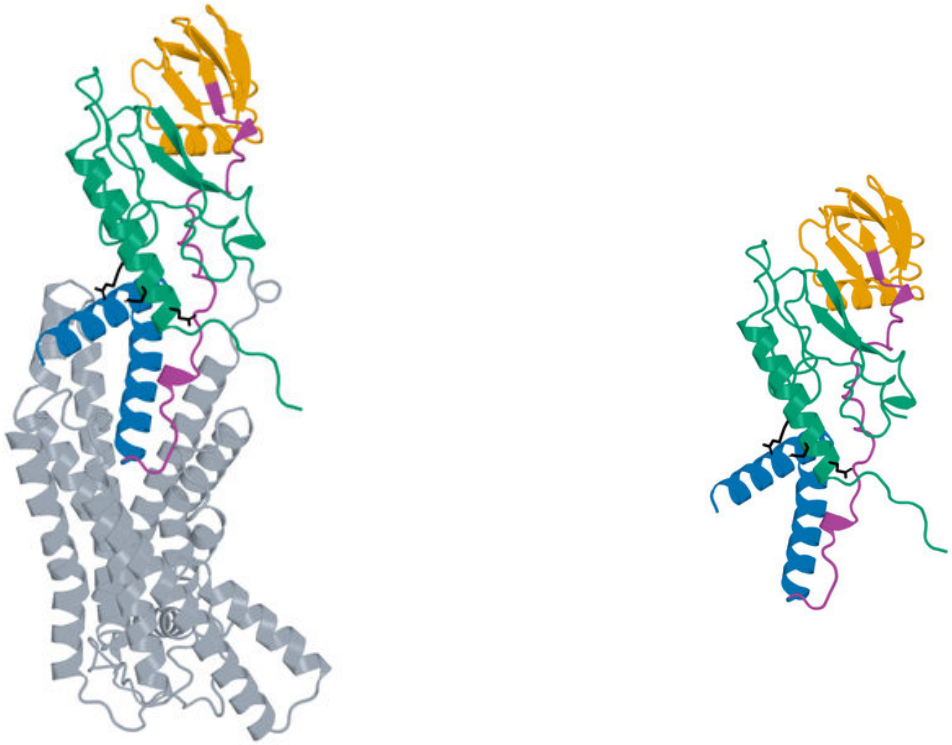


Figure 2. LK_P063_07, final rank 2. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.03 LK_P109_06

Final rank 3 | Two-domain fusion | 113 amino acids

Design history. AP Novo generated the ECD parent S08_00_099 under the condition requesting 20-40-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 64, with ipSAE 0.896. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P109 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 1197 for attempt 6. The allowed linker range was 24-36 residues; the selected linker contains 24 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-20; Linker: residues 21-44; LigandForge domain: residues 45-113. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGRHRVGSGSSPPPPELAPEEVVK
```

Receptor input to linker generation. The retained native receptor intervals were 1-131, 165-199, 202-219, 260-262, 264-266, 268-291, 293-295, 339-345, 348-372, totaling 249 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 374 residues. The gap between the two domains before linker generation was 69.4 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 0, output 99. That batch began with random seed 101; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MSFFQKYLKQLYLQECLSFLGGRHRVGSGSSPPPPELAPEEVVKEVSGATIDLRVKTCVDFKQTVNFF
KLRNTQLPGSCTKDNTVLFLLEGLKLGYASRYFSIGSGSGSQQFS
```

5.03 LK_P109_06 - predicted properties

Boltz2 Exact predicted the full 113-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.33 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.911
Boltz2 Exact	Interface confidence, ipTM	0.871
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.911 / 0.512
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	51.8 / 52.0
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	78.4 / 44.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.419 / 0.289 / 0.439
Design compared with Exact	Fusion DockQ	0.217
Design compared with Exact	Fusion TM-score	0.268
Design compared with Exact	Receptor-core RMSD (Å)	1.4
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	4.7 / 3.3 / 4.3
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	42.1
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	1.33
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	60 / 8 / 7 / 19
Within this design group	Predicted-affinity rank among the scored top 50	12

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.03 LK_P109_06 - bound structures

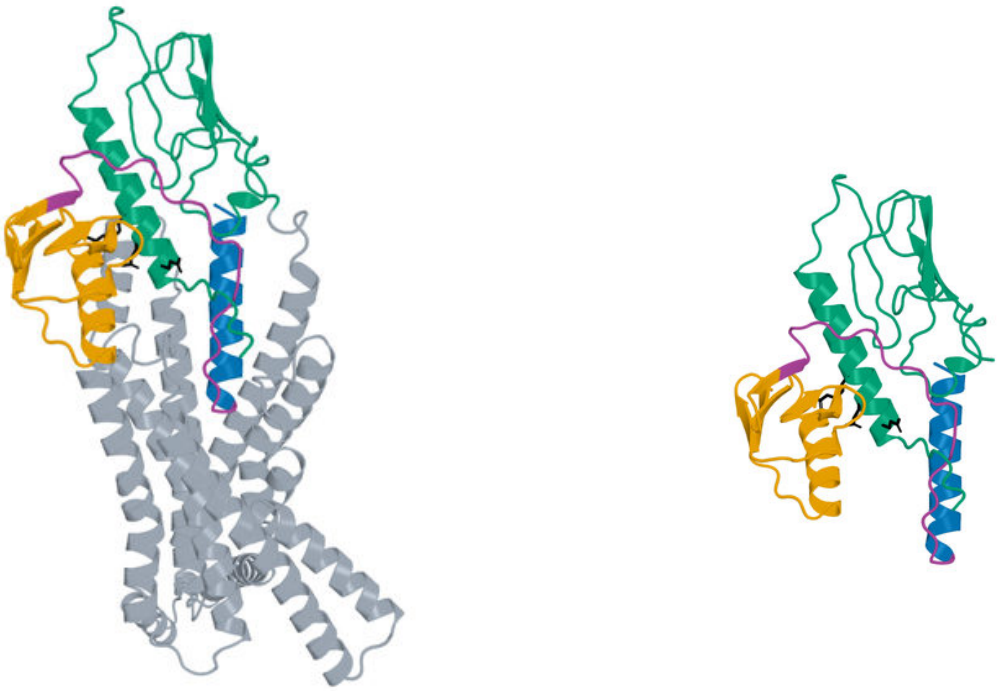


Figure 3. LK_P109_06, final rank 3. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.04 Design_520

Final rank 4 | Original LigandForge design | 53 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 520 in the original 1,000-design population. Its later standalone Exact rank was 4. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.126 (grade C), with MHC-I and MHC-II risk scores of 0.11 and 0.104. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was C, with cleavage-site density 0.483. An N-end-rule proxy estimated 10 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
QDVATDQGILTNVEAIFITNALYFTALLGFSDLCEARNAKKCVGSGSGSGLDR
```

5.04 Design_520 - predicted properties

Boltz2 Exact predicted the full 53-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 3.45 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.911
Boltz2 Exact	Interface confidence, ipTM	0.926
Boltz2 Exact	Mean complex pLDDT (0-1)	0.789
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	3.45
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	83 / 7 / 2 / 13
Within this design group	Predicted-affinity rank among the scored top 50	14

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.04 Design_520 - bound structures

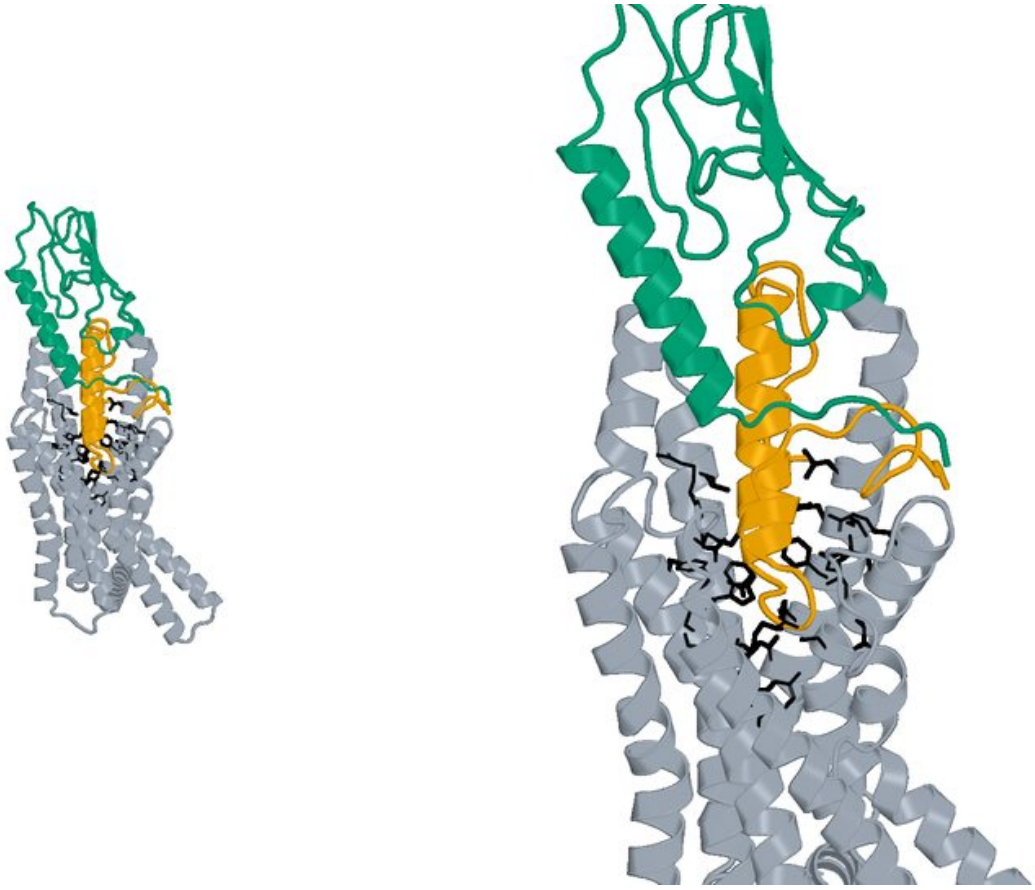


Figure 4. Design_520, final rank 4. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.05 LK_P088_08

Final rank 5 | Two-domain fusion | 153 amino acids

Design history. AP Novo generated the ECD parent S11_03_093 under the condition requesting 40-60-residue binders with receptor positions 106-116 allowed to move. Its standalone Exact rank was 43, with ipSAE 0.902. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P088 was joined in the order LigandForge domain, linker, then ECD domain. AP Novo used random seed 989 for attempt 8. The allowed linker range was 22-34 residues; the selected linker contains 32 residues. Both parent sequences were retained intact.

Sequence map. LigandForge domain: residues 1-69; Linker: residues 70-101; ECD domain: residues 102-153. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
SSSSTVTRRVPPPVNLLPWGPPSGQPFPRGLG
```

Receptor input to linker generation. The retained native receptor intervals were 1-135, 166-199, 202-215, 260-262, 264-266, 268-284, 286-288, 347-370, totaling 233 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 388 residues. The gap between the two domains before linker generation was 64.1 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 3, output 93. That batch began with random seed 401; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
EVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYASRYFSIGSGSGSQQF
SSSSSTVTRRVPPPVNLLPWGPPSGQPFPRGLGMISELQELRRLFESSYPPYFSRLLWSSYLAASVGQ
SDLSPTIGLGKALSASK
```

5.05 LK_P088_08 - predicted properties

Boltz2 Exact predicted the full 153-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 0.718 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.908
Boltz2 Exact	Interface confidence, ipTM	0.883
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.908 / 0.674
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	45.8 / 39.8
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	56.6 / 40.4
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.225 / 0.134 / 0.501
Design compared with Exact	Fusion DockQ	0.139
Design compared with Exact	Fusion TM-score	0.246
Design compared with Exact	Receptor-core RMSD (Å)	0.9
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.0 / 10.0 / 8.4
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	34.0
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	0.718
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	85 / 8 / 3 / 14
Within this design group	Predicted-affinity rank among the scored top 50	1

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.05 LK_P088_08 - bound structures

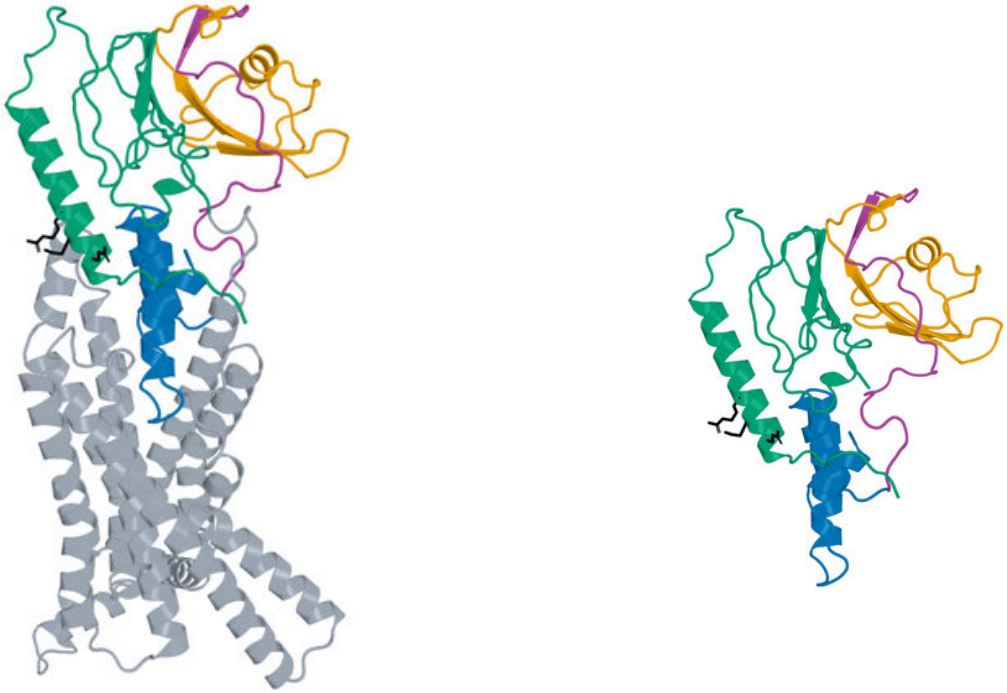


Figure 5. LK_P088_08, final rank 5. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.06 LK_P050_03

Final rank 6 | Two-domain fusion | 149 amino acids

Design history. AP Novo generated the ECD parent S11_02_089 under the condition requesting 40-60-residue binders with receptor positions 106-116 allowed to move. Its standalone Exact rank was 2, with ipSAE 0.927. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P050 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 604 for attempt 3. The allowed linker range was 25-37 residues; the selected linker contains 27 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-53; Linker: residues 54-80; LigandForge domain: residues 81-149. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GFDFVPSPTSSAPPLLCQGTEPAENTR
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 165-222, 253-262, 264-266, 268-296, 337-372, totaling 266 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 425 residues. The gap between the two domains before linker generation was 70.9 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 2, output 89. That batch began with random seed 301; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MGISALEEFDRLRRASILLSLIRSGSTGSGFDAFLEKSGVKGAPAFLAVPSSNGFDFVPSPTSSAPPL
LCQGTEPAENTREVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYASRY
FSIGSGSGSQQFS
```

5.06 LK_P050_03 - predicted properties

Boltz2 Exact predicted the full 149-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 3.28 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.906
Boltz2 Exact	Interface confidence, ipTM	0.920
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.906 / 0.767
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	52.0 / 41.1
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	67.1 / 44.7
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.407 / 0.253 / 0.503
Design compared with Exact	Fusion DockQ	0.207
Design compared with Exact	Fusion TM-score	0.279
Design compared with Exact	Receptor-core RMSD (Å)	0.9
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	3.0 / 3.2 / 3.6
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	38.3
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	3.28
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	69 / 14 / 9 / 20
Within this design group	Predicted-affinity rank among the scored top 50	31

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.06 LK_P050_03 - bound structures

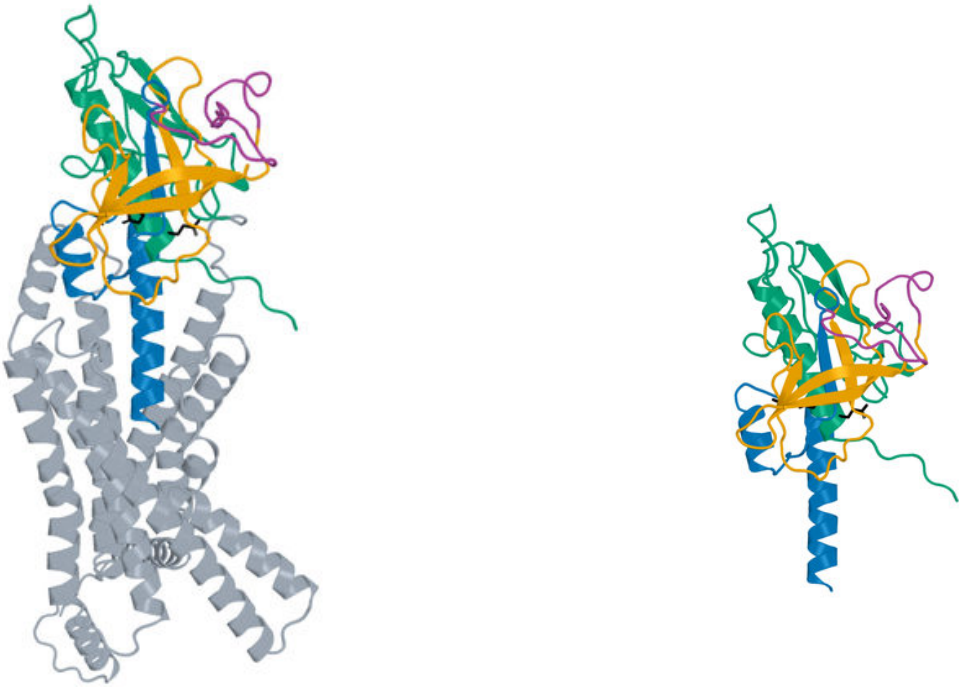


Figure 6. LK_P050_03, final rank 6. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.07 Design_973

Final rank 7 | Original LigandForge design | 65 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 973 in the original 1,000-design population. Its later standalone Exact rank was 8. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.138 (grade C), with MHC-I and MHC-II risk scores of 0.115 and 0.137. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was D, with cleavage-site density 0.523. An N-end-rule proxy estimated 1200 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
SKCTLHALETSFTVDPKGRFKIIAEEFSRTILGCKLLENAKFLYVVGFTEAVNLSGSGSGSPFDR
```

5.07 Design_973 - predicted properties

Boltz2 Exact predicted the full 65-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 0.619 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.904
Boltz2 Exact	Interface confidence, ipTM	0.948
Boltz2 Exact	Mean complex pLDDT (0-1)	0.768
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	0.619
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	76 / 8 / 3 / 20
Within this design group	Predicted-affinity rank among the scored top 50	1

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.07 Design_973 - bound structures

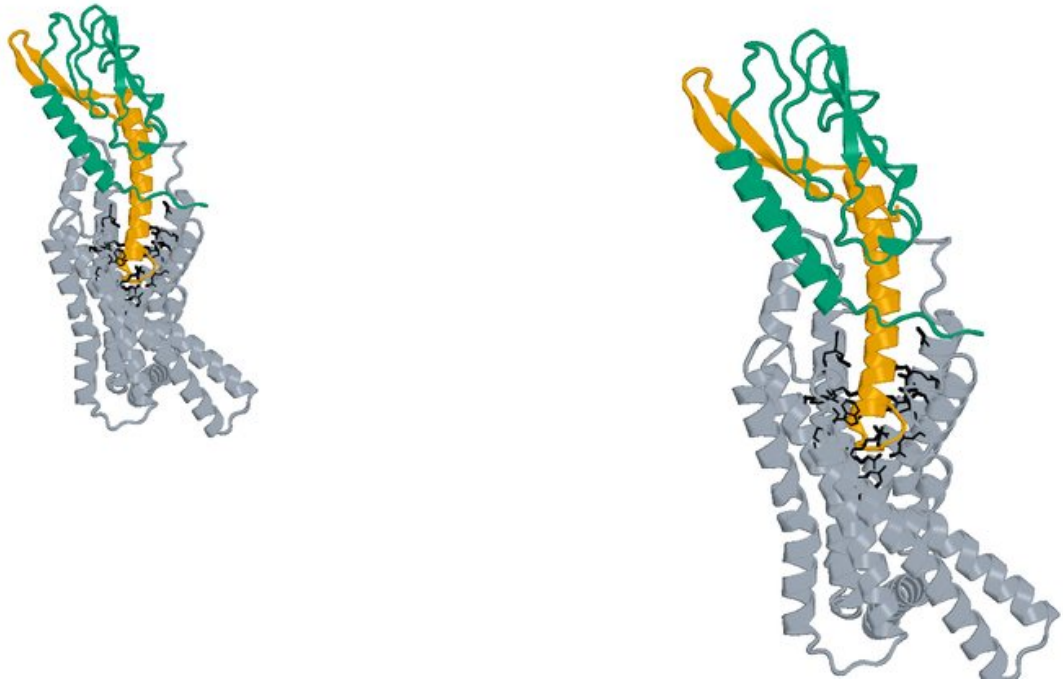


Figure 7. Design_973, final rank 7. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.08 LK_P157_08

Final rank 8 | Two-domain fusion | 139 amino acids

Design history. AP Novo generated the ECD parent S10_02_023 under the condition requesting 40-60-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 21, with ipSAE 0.907. LigandForge generated the other parent, Design_414, which had standalone Exact rank 990 and ipSAE 0.000. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P157 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 1679 for attempt 8. The allowed linker range was 24-36 residues; the selected linker contains 35 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-50; Linker: residues 51-85; LigandForge domain: residues 86-139. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGTDSLGKGSYDDKLNALPFSPEAEAAAQKDVEAG
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 163-200, 202-222, 256-262, 264-266, 268-296, 337-372, totaling 264 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 404 residues. The gap between the two domains before linker generation was 67.6 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 2, output 23. That batch began with random seed 301; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MSIDYSSLVTHSPTGVAEATEFLYRYYQLREIPLETFIAVASALYHSSLLGGTDSLGKGSYDDKLNAL
PFSPEAEAAAQKDVEAGKTKEGSSCVRGGSRTELFSKEPDKGVSFVSGSGVEYVCFAFFVVGSGSGST
YVS
```

5.08 LK_P157_08 - predicted properties

Boltz2 Exact predicted the full 139-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 3.24 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.902
Boltz2 Exact	Interface confidence, ipTM	0.888
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.902 / 0.596
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	54.2 / 46.3
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	57.5 / 56.2
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.315 / 0.131 / 0.529
Design compared with Exact	Fusion DockQ	0.155
Design compared with Exact	Fusion TM-score	0.256
Design compared with Exact	Receptor-core RMSD (Å)	1.2
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.7 / 5.2 / 6.6
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	76.8
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.4 / 0
DeltaForge	Predicted Kd (nM)	3.24
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	120 / 12 / 3 / 27
Within this design group	Predicted-affinity rank among the scored top 50	30

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.08 LK_P157_08 - bound structures

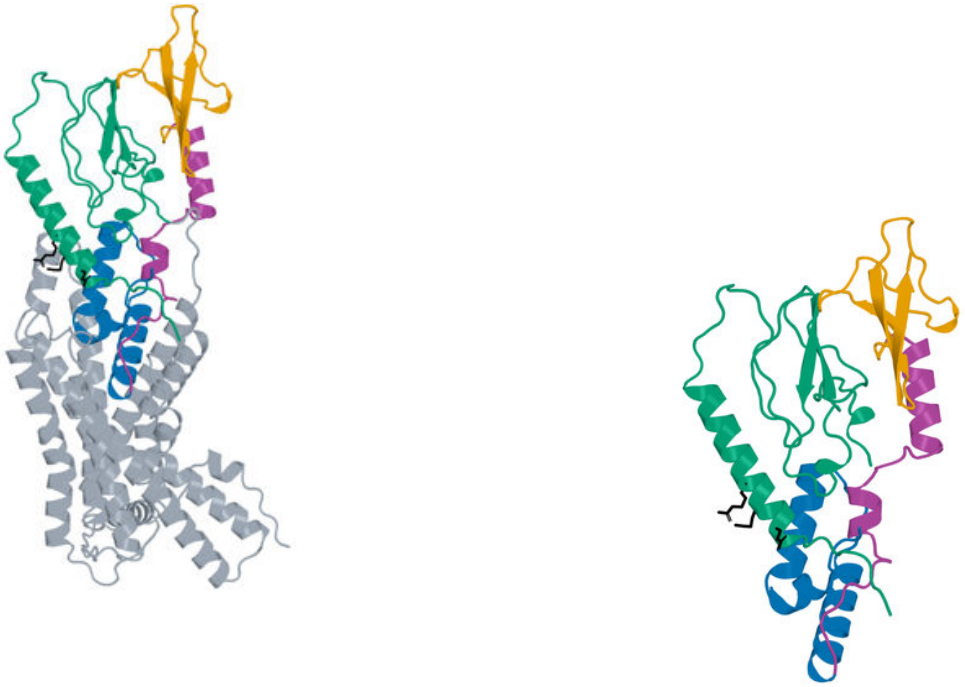


Figure 8. LK_P157_08, final rank 8. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.09 LK_P069_04

Final rank 9 | Two-domain fusion | 154 amino acids

Design history. AP Novo generated the ECD parent S10_02_023 under the condition requesting 40-60-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 21, with ipSAE 0.907. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P069 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 795 for attempt 4. The allowed linker range was 25-37 residues; the selected linker contains 35 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-50; Linker: residues 51-85; LigandForge domain: residues 86-154. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGLGEFSDNGLPPVAPLPPGLPVPDLPQDEESDIK
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 163-200, 202-222, 256-262, 264-266, 268-296, 337-372, totaling 264 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 420 residues. The gap between the two domains before linker generation was 72.2 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 2, output 23. That batch began with random seed 301; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MSIDYSSLVTHSPTGVAEATEFLYRYYQLREIPLETFIAVASALYHSSLLGGLGEFSDNGLPPVAPLP
PGLPVPDLPQDEESDIKEVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLG
YASRYFSIGSGSGSQQFS
```

5.09 LK_P069_04 - predicted properties

Boltz2 Exact predicted the full 154-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.25 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.902
Boltz2 Exact	Interface confidence, ipTM	0.897
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.902 / 0.690
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	47.1 / 40.5
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	61.4 / 40.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.244 / 0.158 / 0.508
Design compared with Exact	Fusion DockQ	0.12
Design compared with Exact	Fusion TM-score	0.36
Design compared with Exact	Receptor-core RMSD (Å)	1.1
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	4.8 / 3.5 / 2.4
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	48.5
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	1.25
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	89 / 18 / 7 / 27
Within this design group	Predicted-affinity rank among the scored top 50	11

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.09 LK_P069_04 - bound structures

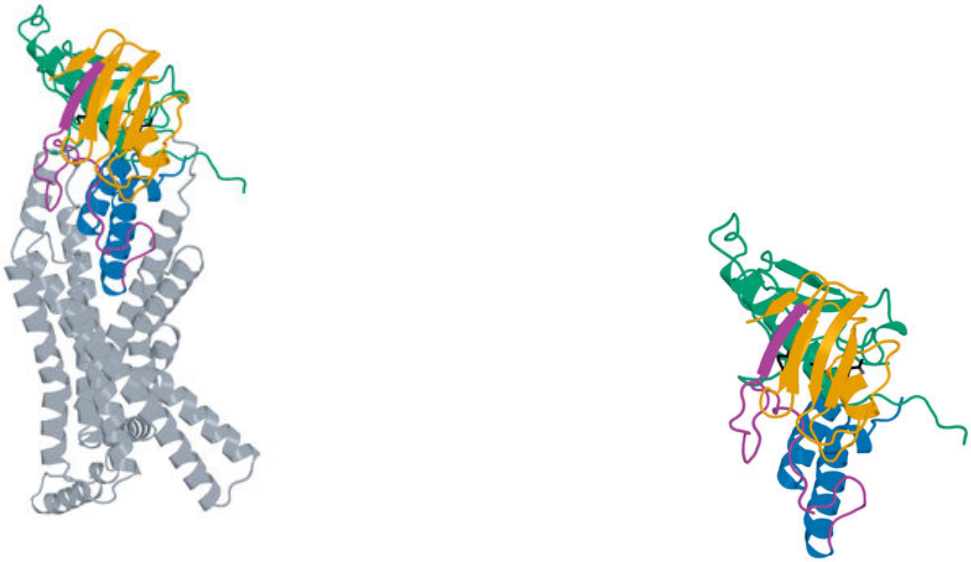


Figure 9. LK_P069_04, final rank 9. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.10 LK_P024_03

Final rank 10 | Two-domain fusion | 128 amino acids

Design history. AP Novo generated the ECD parent S11_00_040 under the condition requesting 40-60-residue binders with receptor positions 106-116 allowed to move. Its standalone Exact rank was 17, with ipSAE 0.910. LigandForge generated the other parent, Design_613, which had standalone Exact rank 970 and ipSAE 0.012. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P024 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 344 for attempt 3. The allowed linker range was 16-28 residues; the selected linker contains 18 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-48; Linker: residues 49-66; LigandForge domain: residues 67-128. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGHRVEPPVRPPPVSLPG
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 165-219, 260-262, 264-266, 268-291, 340-370, totaling 246 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 384 residues. The gap between the two domains before linker generation was 46.7 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 0, output 40. That batch began with random seed 101; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MRALAQASTGSSALETLHRMRVEAFFQQQLRSGTLSDGAVEAYSRLVGGGHRVEPPVRPPPVSLPGRT
CTSAYGVPYSDGHFQRPVPSENGLNTQGVTDVSGRETDESTDVFVVISCQGSGSGSTRFN
```

5.10 LK_P024_03 - predicted properties

Boltz2 Exact predicted the full 128-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 3.79 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.902
Boltz2 Exact	Interface confidence, ipTM	0.871
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.902 / 0.561
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	57.3 / 51.5
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	69.3 / 49.8
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.338 / 0.232 / 0.467
Design compared with Exact	Fusion DockQ	0.214
Design compared with Exact	Fusion TM-score	0.351
Design compared with Exact	Receptor-core RMSD (Å)	0.9
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	6.0 / 6.5 / 2.9
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	39.3
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	3.79
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	65 / 11 / 8 / 16
Within this design group	Predicted-affinity rank among the scored top 50	33

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.10 LK_P024_03 - bound structures

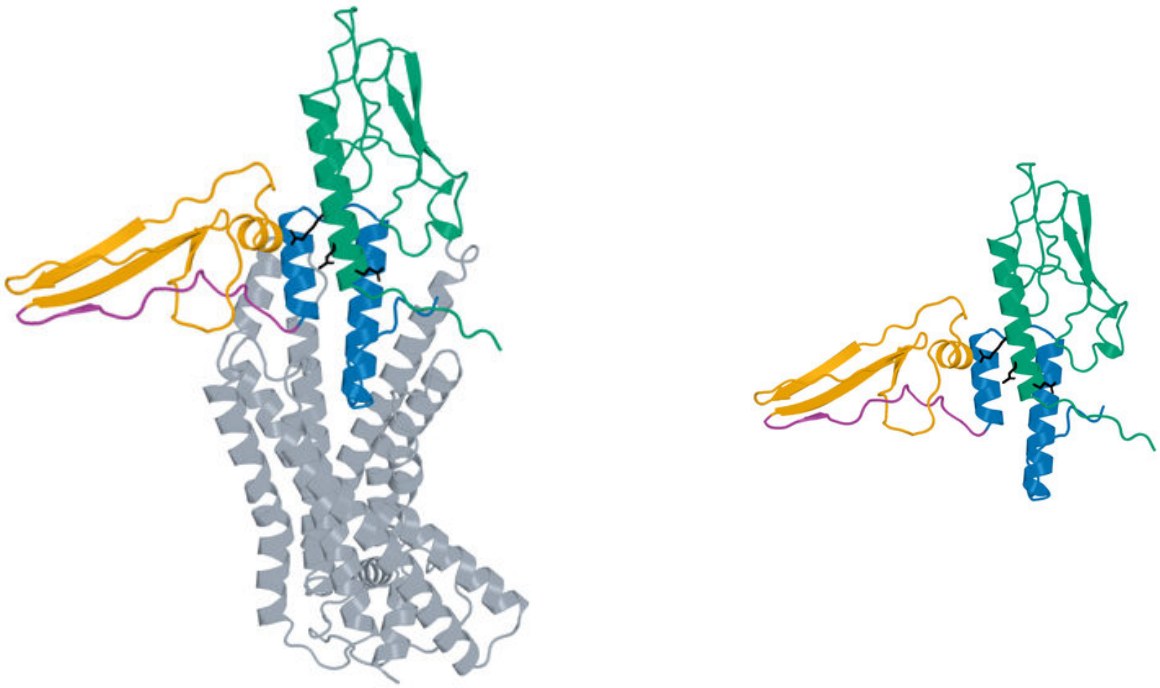


Figure 10. LK_P024_03, final rank 10. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.11 Design_994

Final rank 11 | Original LigandForge design | 70 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 994 in the original 1,000-design population. Its later standalone Exact rank was 11. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.097 (grade B), with MHC-I and MHC-II risk scores of 0.1 and 0.057. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was C, with cleavage-site density 0.489. An N-end-rule proxy estimated 420 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
TIDGYKHLFEISDVIEAEGGELQLKRQEYESAIGFKLCRLNAESVYNVKKLSGLRPVGCKGSGSGSYV
NL
```

5.11 Design_994 - predicted properties

Boltz2 Exact predicted the full 70-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 2.74 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.901
Boltz2 Exact	Interface confidence, ipTM	0.955
Boltz2 Exact	Mean complex pLDDT (0-1)	0.744
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	2.74
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	87 / 13 / 5 / 13
Within this design group	Predicted-affinity rank among the scored top 50	12

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.11 Design_994 - bound structures

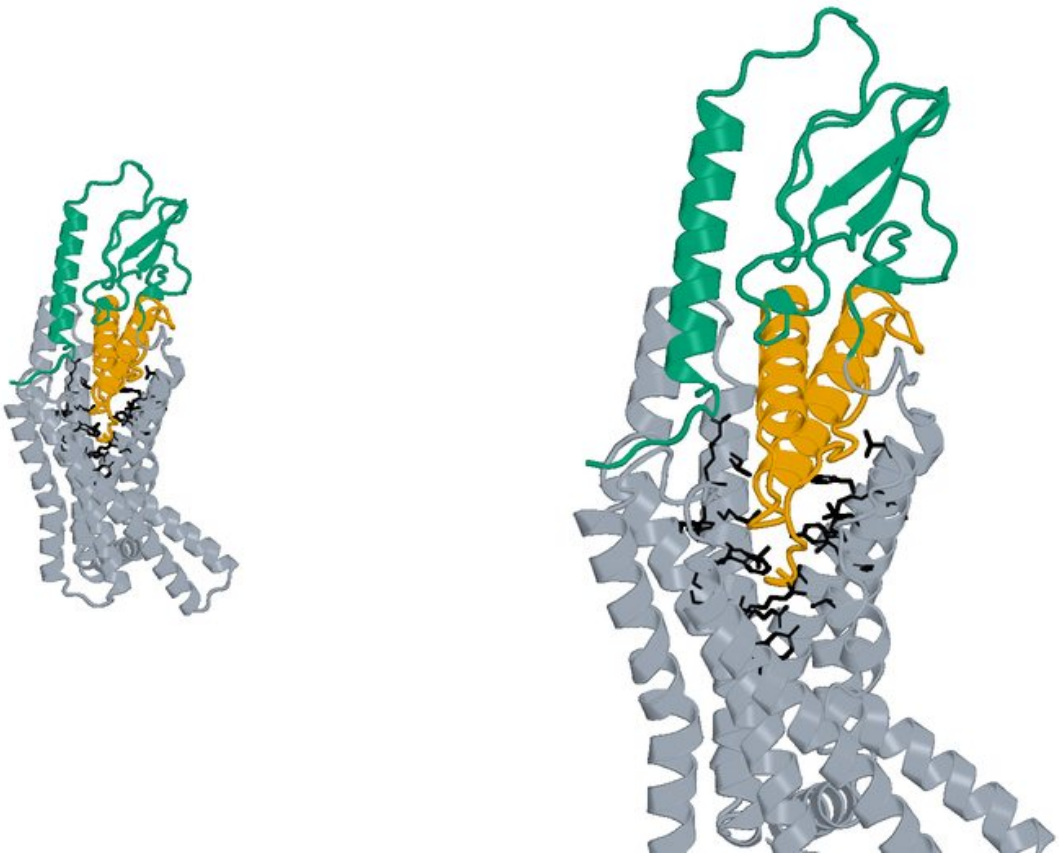


Figure 11. Design_994, final rank 11. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.12 LK_P012_08

Final rank 12 | Two-domain fusion | 125 amino acids

Design history. AP Novo generated the ECD parent S08_00_042 under the condition requesting 20-40-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 40, with ipSAE 0.903. LigandForge generated the other parent, Design_991, which had standalone Exact rank 947 and ipSAE 0.074. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P012 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 229 for attempt 8. The allowed linker range was 15-27 residues; the selected linker contains 21 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-39; Linker: residues 40-60; LigandForge domain: residues 61-125. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGAPSFLPPPNVTSGYVNGGG
```

Receptor input to linker generation. The retained native receptor intervals were 1-106, 109-130, 163-219, 256-262, 264-295, 339-372, totaling 258 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 389 residues. The gap between the two domains before linker generation was 42.9 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 0, output 42. That batch began with random seed 101; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MADSISEVSEGVHLILAAQALELPPFVVAARLEEARAKRGGAPSFLPPPNVTSGYVNGGGAFHLTGLA
ARCGAVWRGVFCQFGNGVKGEGSVTGGSLNTFYILTVAPTSANILTSGSGSGSLEGS
```

5.12 LK_P012_08 - predicted properties

Boltz2 Exact predicted the full 125-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.44 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.901
Boltz2 Exact	Interface confidence, ipTM	0.918
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.901 / 0.740
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	47.9 / 41.6
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	67.4 / 38.2
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.235 / 0.112 / 0.554
Design compared with Exact	Fusion DockQ	0.216
Design compared with Exact	Fusion TM-score	0.289
Design compared with Exact	Receptor-core RMSD (Å)	0.9
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	6.4 / 5.1 / 3.1
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	34.0
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	1.44
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	114 / 10 / 9 / 32
Within this design group	Predicted-affinity rank among the scored top 50	14

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.12 LK_P012_08 - bound structures

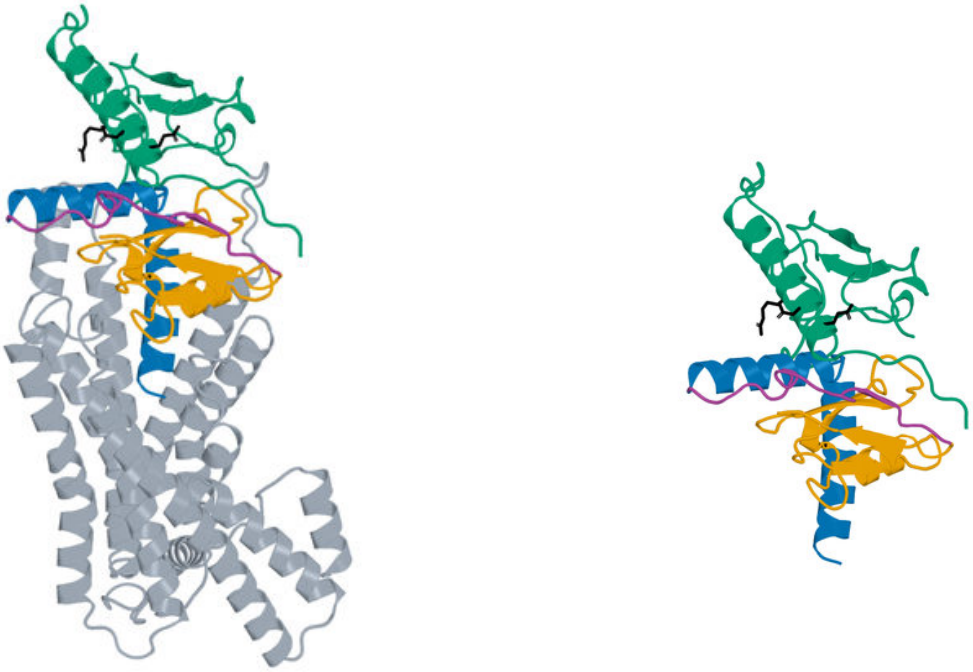


Figure 12. LK_P012_08, final rank 12. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.13 LK_P062_00

Final rank 13 | Two-domain fusion | 146 amino acids

Design history. AP Novo generated the ECD parent S10_04_045 under the condition requesting 40-60-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 14, with ipSAE 0.911. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P062 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 721 for attempt 0. The allowed linker range was 24-36 residues; the selected linker contains 31 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-46; Linker: residues 47-77; LigandForge domain: residues 78-146. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GGVGSGHPLEPITIVRPPPVPELEPITGPVT
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 163-200, 202-222, 260-262, 264-266, 268-295, 337-372, totaling 259 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 410 residues. The gap between the two domains before linker generation was 67.6 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 4, output 45. That batch began with random seed 501; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MTDIGEEIKALVALTEGVPPENRPFAVELAQRLSPWFSLYSRLSSRGGVGSGHPLEPITIVRPPPVPE
LEPITGPVTEVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYASRYFSI
GSGSGSQQFS
```

5.13 LK_P062_00 - predicted properties

Boltz2 Exact predicted the full 146-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.13 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.900
Boltz2 Exact	Interface confidence, ipTM	0.878
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.900 / 0.708
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	55.9 / 49.4
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	61.5 / 55.0
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.258 / 0.125 / 0.602
Design compared with Exact	Fusion DockQ	0.105
Design compared with Exact	Fusion TM-score	0.265
Design compared with Exact	Receptor-core RMSD (Å)	1.8
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.7 / 6.2 / 4.2
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	49.7
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	1.13
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	83 / 6 / 5 / 17
Within this design group	Predicted-affinity rank among the scored top 50	10

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.13 LK_P062_00 - bound structures

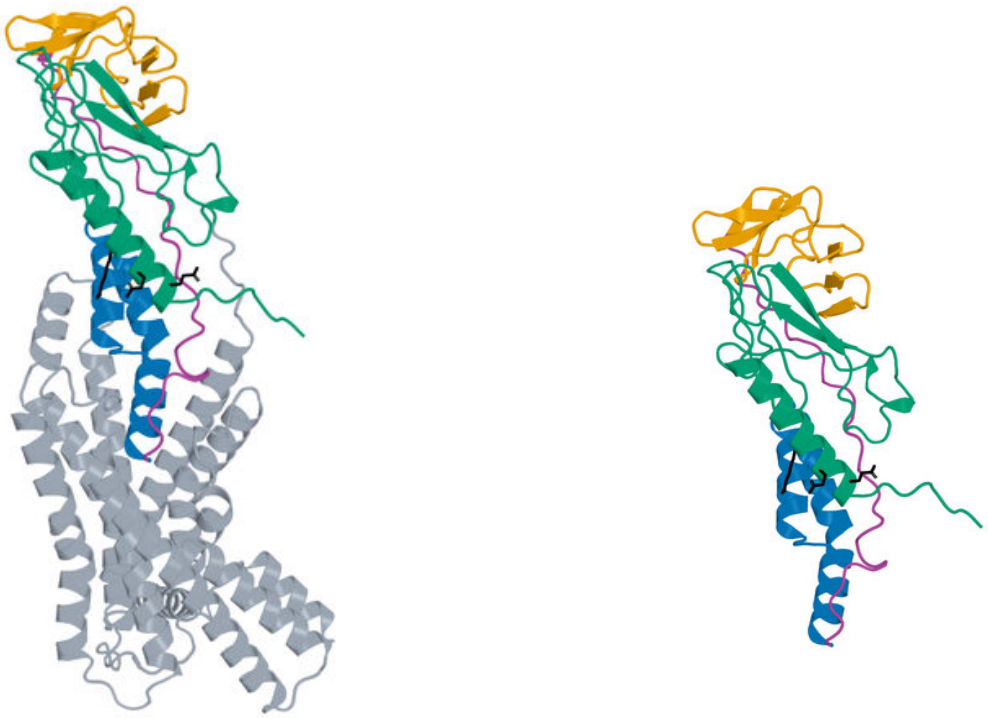


Figure 13. LK_P062_00, final rank 13. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.14 Design_440

Final rank 14 | Original LigandForge design | 54 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 440 in the original 1,000-design population. Its later standalone Exact rank was 13. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.141 (grade C), with MHC-I and MHC-II risk scores of 0.14 and 0.102. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was D, with cleavage-site density 0.519. An N-end-rule proxy estimated 7200 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
PNYFGFTGFLCIVGDINIGRKAEEFQFFGAIKGGGIEFVCKSDVGSGSGSILFN
```

5.14 Design_440 - predicted properties

Boltz2 Exact predicted the full 54-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 4.4 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.900
Boltz2 Exact	Interface confidence, ipTM	0.963
Boltz2 Exact	Mean complex pLDDT (0-1)	0.790
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	4.4
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	95 / 9 / 5 / 21
Within this design group	Predicted-affinity rank among the scored top 50	19

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.14 Design_440 - bound structures

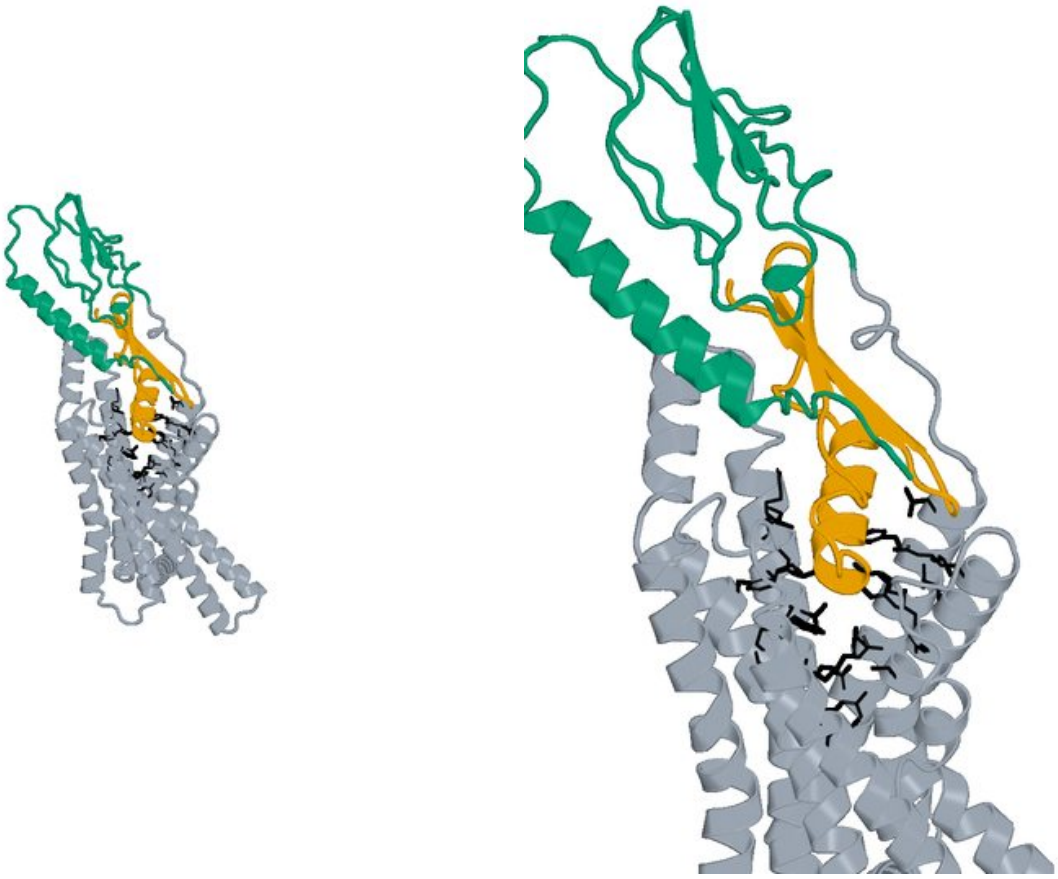


Figure 14. Design_440, final rank 14. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.15 LK_P141_07

Final rank 15 | Two-domain fusion | 121 amino acids

Design history. AP Novo generated the ECD parent S09_03_086 under the condition requesting 20-40-residue binders with receptor positions 106-116 allowed to move. Its standalone Exact rank was 97, with ipSAE 0.891. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P141 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 1518 for attempt 7. The allowed linker range was 16-28 residues; the selected linker contains 20 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-32; Linker: residues 33-52; LigandForge domain: residues 53-121. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GSGLPPPPPPVDVIPPSEVK
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 165-222, 256-262, 264-266, 268-296, 337-372, totaling 263 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 392 residues. The gap between the two domains before linker generation was 45.0 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 3, output 86. That batch began with random seed 401; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MVSIGSTGFDGLLQLDEELRNLFLQRLVLGRLGSGLPPPPPPVDVIPPSEVKEVSGATIDLRVKTCVD
FKQTVNFFKLRNTQLPGSCTKDNTVLFLLEGLKLGYASRYFSIGSGSGSQQFS
```

5.15 LK_P141_07 - predicted properties

Boltz2 Exact predicted the full 121-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 1.63 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.900
Boltz2 Exact	Interface confidence, ipTM	0.884
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.900 / 0.483
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	59.9 / 61.7
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	71.4 / 54.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.412 / 0.265 / 0.469
Design compared with Exact	Fusion DockQ	0.274
Design compared with Exact	Fusion TM-score	0.301
Design compared with Exact	Receptor-core RMSD (Å)	0.6
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.4 / 10.7 / 8.8
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	76.4
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	1.63
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	62 / 7 / 5 / 18
Within this design group	Predicted-affinity rank among the scored top 50	17

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.15 LK_P141_07 - bound structures

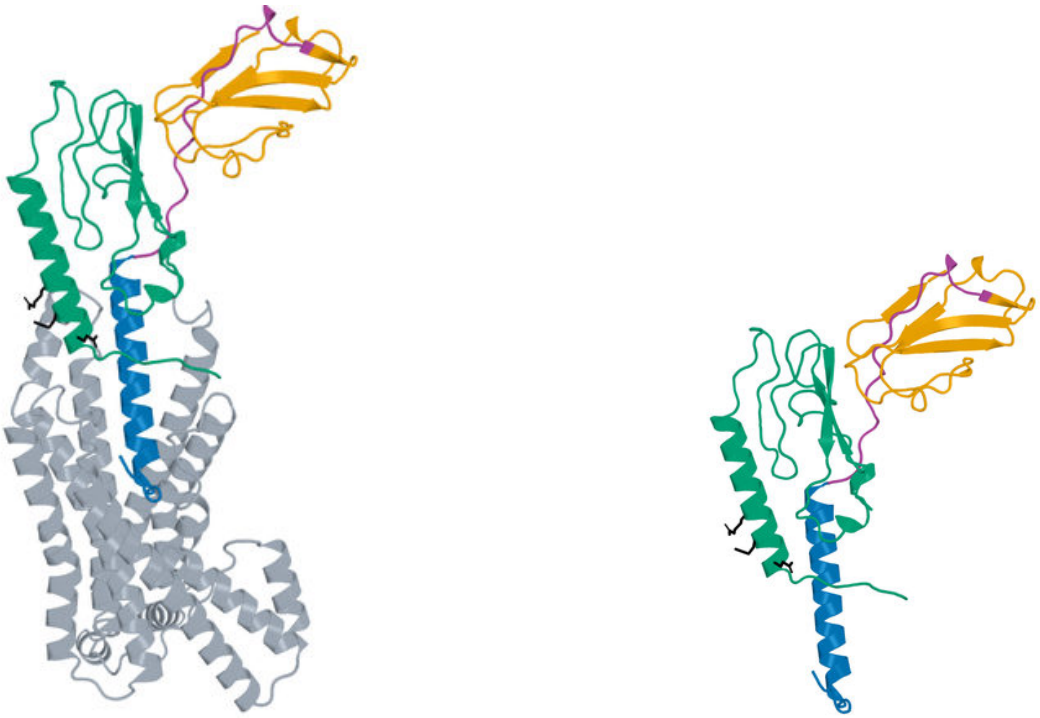


Figure 15. LK_P141_07, final rank 15. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.16 LK_P002_05

Final rank 16 | Two-domain fusion | 104 amino acids

Design history. AP Novo generated the ECD parent S10_03_014 under the condition requesting 40-60-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 6, with ipSAE 0.920. LigandForge generated the other parent, Design_913, which had standalone Exact rank 859 and ipSAE 0.422. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P002 was joined in the order LigandForge domain, linker, then ECD domain. AP Novo used random seed 126 for attempt 5. The allowed linker range was 18-30 residues; the selected linker contains 21 residues. Both parent sequences were retained intact.

Sequence map. LigandForge domain: residues 1-36; Linker: residues 37-57; ECD domain: residues 58-104. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GKTSTSGPQPVIILPVPVLPG
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 142-248, 256-262, 264-266, 268-296, 299-393, totaling 371 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 484 residues. The gap between the two domains before linker generation was 52.2 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 3, output 14. That batch began with random seed 401; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
NLPRIECSSNKGNVQEDSDVIGGRGSGSGSGSCGIFGKTSTSGPQPVIILPVPVLPGMLPQLFLTHML
SVMVGTGSLDSGLVSAFRGSPLVSEAILGFLKQYLS
```

5.16 LK_P002_05 - predicted properties

Boltz2 Exact predicted the full 104-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 0.813 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.899
Boltz2 Exact	Interface confidence, ipTM	0.937
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.899 / 0.779
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	43.8 / 38.2
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	51.5 / 37.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.183 / 0.101 / 0.617
Design compared with Exact	Fusion DockQ	0.0893
Design compared with Exact	Fusion TM-score	0.419
Design compared with Exact	Receptor-core RMSD (Å)	0.9
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	8.2 / 5.6 / 2.7
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	42.1
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	0.813
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	97 / 5 / 0 / 19
Within this design group	Predicted-affinity rank among the scored top 50	4

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.16 LK_P002_05 - bound structures

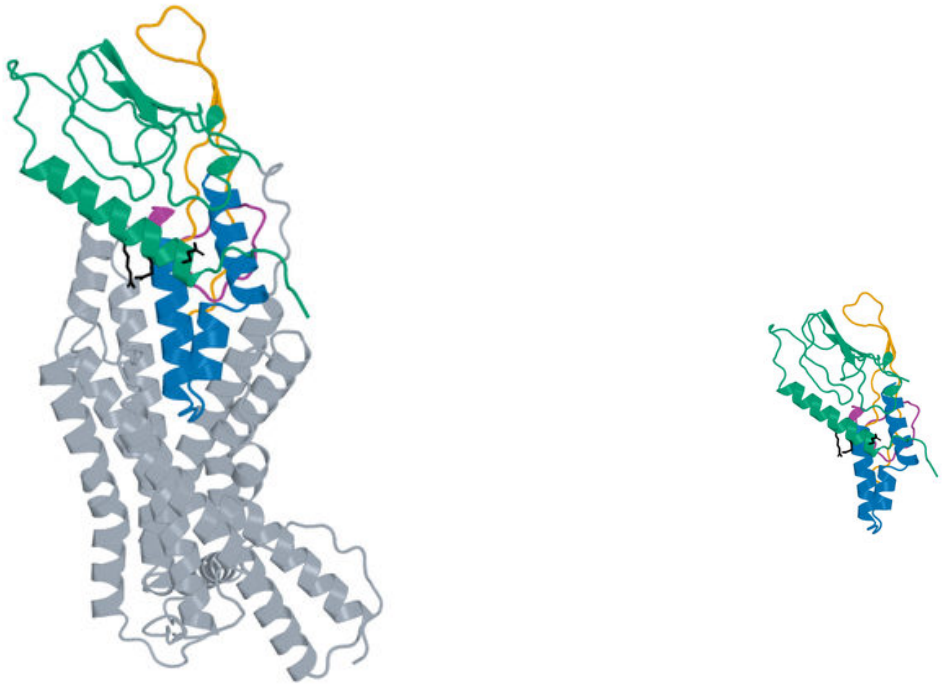


Figure 16. LK_P002_05, final rank 16. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.17 LK_P012_04

Final rank 17 | Two-domain fusion | 130 amino acids

Design history. AP Novo generated the ECD parent S08_00_042 under the condition requesting 20-40-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 40, with ipSAE 0.903. LigandForge generated the other parent, Design_991, which had standalone Exact rank 947 and ipSAE 0.074. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P012 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 225 for attempt 4. The allowed linker range was 15-27 residues; the selected linker contains 26 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-39; Linker: residues 40-65; LigandForge domain: residues 66-130. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
PGSSLLPLPEDTSDVASGSISYPGAS
```

Receptor input to linker generation. The retained native receptor intervals were 1-106, 109-130, 163-219, 256-262, 264-295, 339-372, totaling 258 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 389 residues. The gap between the two domains before linker generation was 42.9 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 0, output 42. That batch began with random seed 101; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
MADSISEVSEGVHLILAAQALELPPFVVAARLEEARAKRPGSSLLPLPEDTSDVASGSISYPGASAFH
LTGLAARCGAVWRGVFCQFGNGVKGEGSVTGGSLNTFYILTVAPTSANILTSGSGSGSLEGS
```

5.17 LK_P012_04 - predicted properties

Boltz2 Exact predicted the full 130-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 2.15 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.899
Boltz2 Exact	Interface confidence, ipTM	0.885
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.899 / 0.649
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	46.2 / 36.0
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	70.6 / 35.7
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.245 / 0.113 / 0.524
Design compared with Exact	Fusion DockQ	0.218
Design compared with Exact	Fusion TM-score	0.362
Design compared with Exact	Receptor-core RMSD (Å)	0.8
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	7.1 / 3.1 / 2.9
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	33.8
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	2.15
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	98 / 18 / 9 / 27
Within this design group	Predicted-affinity rank among the scored top 50	22

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

5.17 LK_P012_04 - bound structures

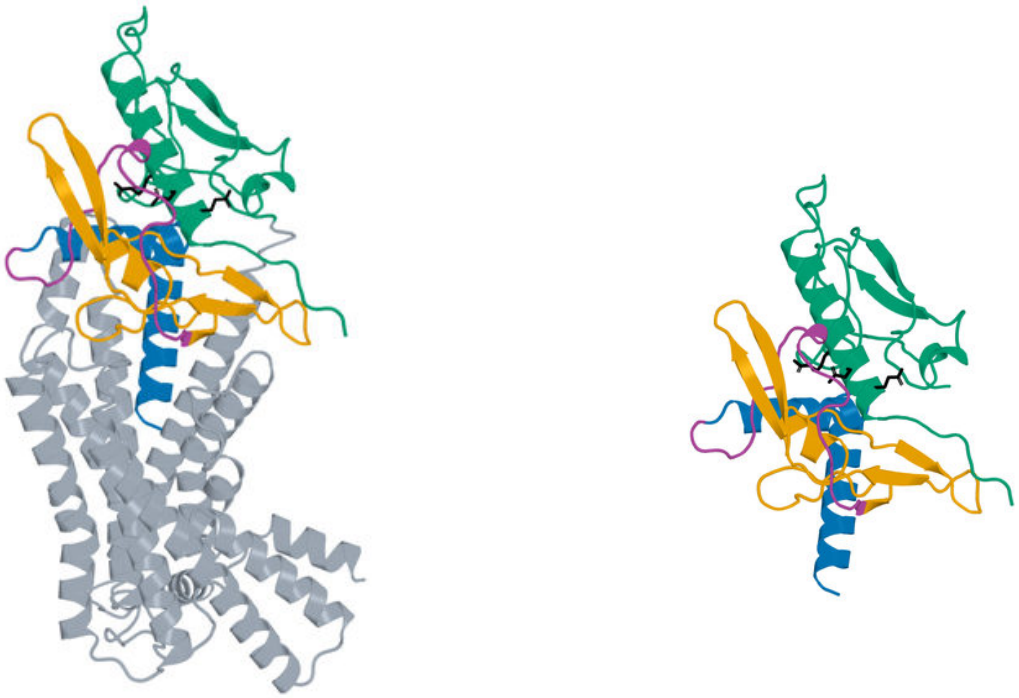


Figure 17. LK_P012_04, final rank 17. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.18 Design_787

Final rank 18 | Original LigandForge design | 49 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 787 in the original 1,000-design population. Its later standalone Exact rank was 14. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.061 (grade B), with MHC-I and MHC-II risk scores of 0.037 and 0.076. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was C, with cleavage-site density 0.478. An N-end-rule proxy estimated 1800 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
GTLALTHCVFAADRAEIFGQRGKEDLQTGKAASEGTELTGSGSGSLCDI
```

5.18 Design_787 - predicted properties

Boltz2 Exact predicted the full 49-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 5.38 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.899
Boltz2 Exact	Interface confidence, ipTM	0.955
Boltz2 Exact	Mean complex pLDDT (0-1)	0.786
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	5.38
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	100 / 7 / 6 / 20
Within this design group	Predicted-affinity rank among the scored top 50	20

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.18 Design_787 - bound structures

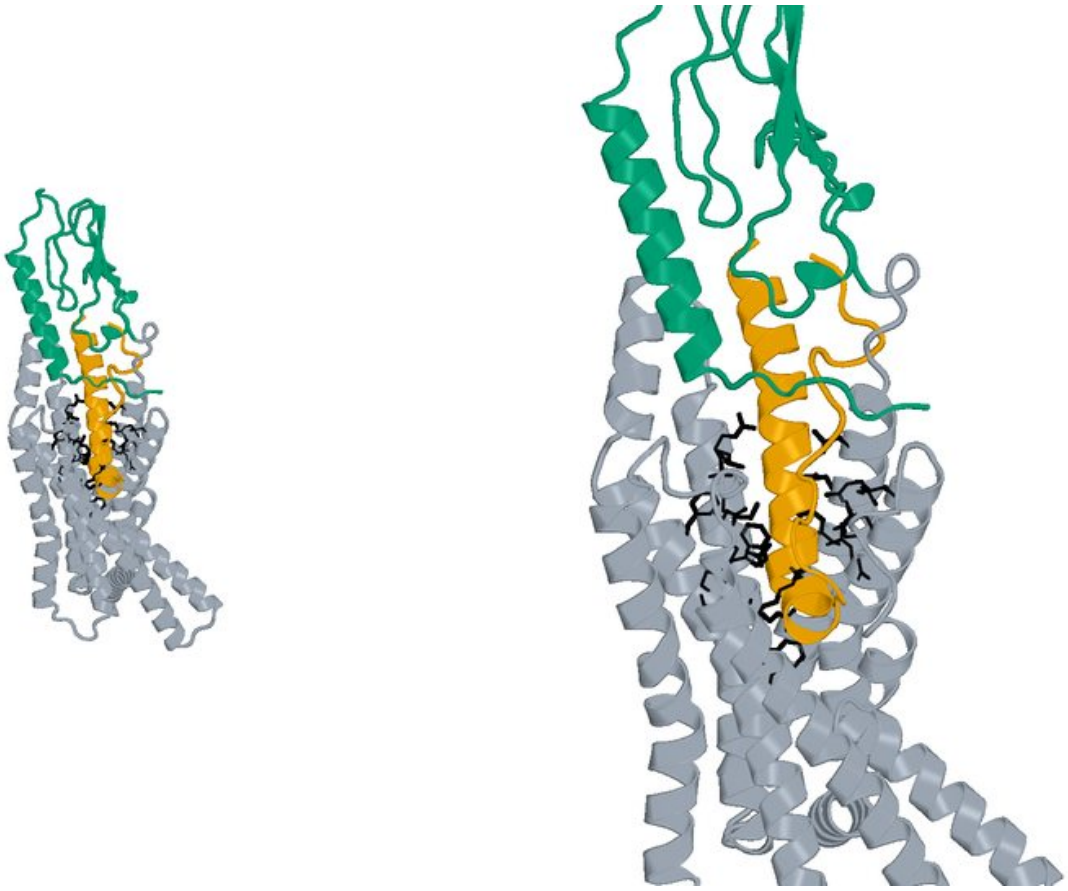


Figure 18. Design_787, final rank 18. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.19 Design_15

Final rank 19 | Original LigandForge design | 50 amino acids

Design history. LigandForge version 6.5.21 produced this sequence as output 15 in the original 1,000-design population. Its later standalone Exact rank was 15. The full exported sequence, including the fixed GSGSGS region and final four residues, was retained. It was not fused, shortened or redesigned. The original target-pocket uncertainty described in section 2.1 applies.

Inputs and validation. Boltz2 Exact received the complete candidate sequence and the 409-residue receptor sequence. Prediction used seed 101, three recycling steps, 200 sampling steps and one diffusion sample, without a template or hotspot constraint. DeltaForge then scored that saved complex. Sequence and input checks linked the affinity result to this exact molecule. The original LigandForge complex coordinates were not retained, so structural agreement with the generated design cannot be assessed.

Additional generator estimates. The provider reported immunogenicity risk 0.122 (grade C), with MHC-I and MHC-II risk scores of 0.102 and 0.112. MHC refers to the major histocompatibility complex involved in immune presentation. The predicted proteolytic grade was D, with cleavage-site density 0.55. An N-end-rule proxy estimated 4200 minutes; it is not a measured half-life. These provider estimates have no validation or uncertainty intervals in this study.

Complete selected sequence, from N terminus to C terminus:

```
AQCTKTYAIGYDVYLKLTYLGFRRADGAGNDDDVCAGNVGGSGSGSFFVA
```

5.19 Design_15 - predicted properties

Boltz2 Exact predicted the full 50-residue candidate against the 409-residue receptor using seed 101. DeltaForge predicted Kd = 4.29 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.899
Boltz2 Exact	Interface confidence, ipTM	0.947
Boltz2 Exact	Mean complex pLDDT (0-1)	0.694
Generated design compared with Exact	Structural agreement	Unavailable: original generated coordinates missing
DeltaForge	Predicted Kd (nM)	4.29
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	74 / 8 / 4 / 17
Within this design group	Predicted-affinity rank among the scored top 50	18

Interpretation. The interface scores and predicted affinity support experimental prioritization. They do not resolve the original target-pocket uncertainty or establish receptor activation.

5.19 Design_15 - bound structures

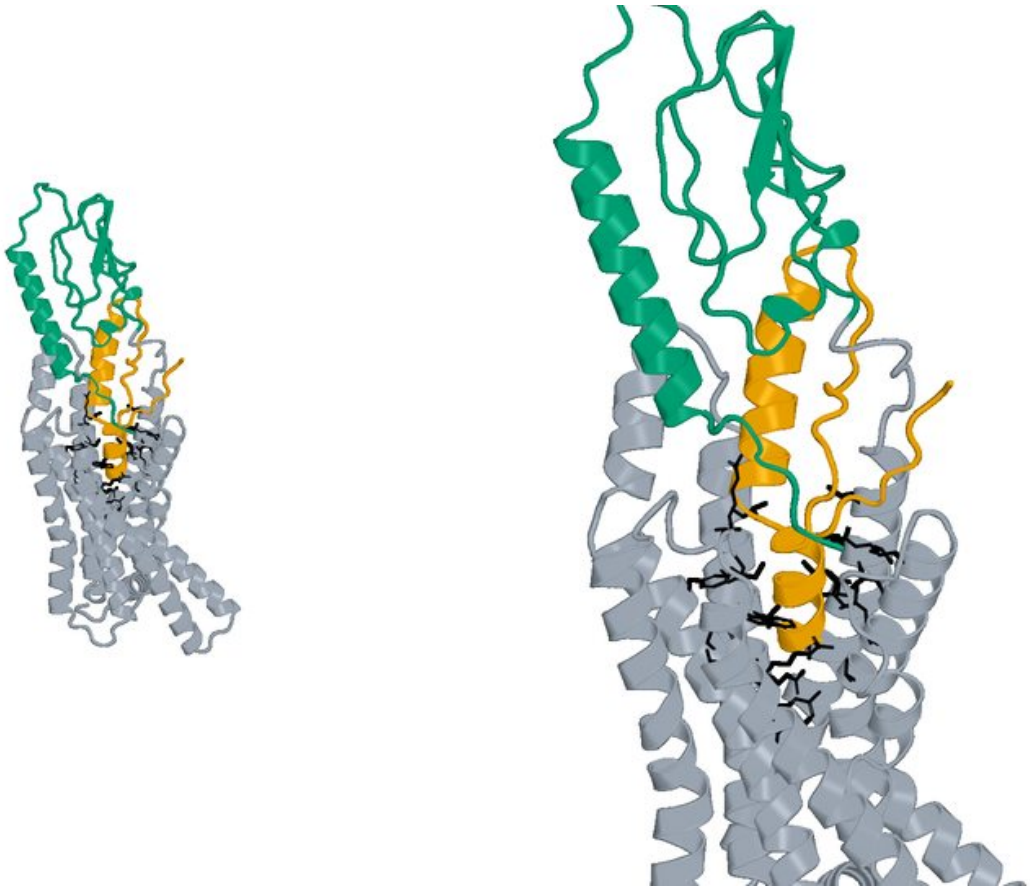


Figure 19. Design_15, final rank 19. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 409-residue receptor as chain B. Left: full complex. Right: interface detail. The complete LigandForge candidate is orange. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show the 27 originally requested pocket positions. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

5.20 LK_P093_03

Final rank 20 | Two-domain fusion | 159 amino acids

Design history. AP Novo generated the ECD parent S10_02_067 under the condition requesting 40-60-residue binders with all receptor coordinates fixed. Its standalone Exact rank was 48, with ipSAE 0.900. LigandForge generated the other parent, Design_280, which had standalone Exact rank 979 and ipSAE 0.011. These scores describe each parent alone, before fusion.

How the fusion was made. Parent pair P093 was joined in the order ECD domain, linker, then LigandForge domain. AP Novo used random seed 1034 for attempt 3. The allowed linker range was 23-35 residues; the selected linker contains 30 residues. Both parent sequences were retained intact.

Sequence map. ECD domain: residues 1-60; Linker: residues 61-90; LigandForge domain: residues 91-159. Positions count from the first amino acid of the complete sequence below.

Generated linker sequence:

```
GISSSVPDPRTPPPPPPPVPVEPLDPSSLT
```

Receptor input to linker generation. The retained native receptor intervals were 1-130, 166-200, 202-215, 217-219, 260-262, 264-266, 268-291, 340-370, totaling 243 residues. The context radius was 10.0 Å. The largest input allowed by this pair's linker range contained 407 residues. The gap between the two domains before linker generation was 65.2 Å. The fixed domains retained their input geometry, with C-alpha RMSD below 0.001 Å during generation. The complete 440-residue receptor was then restored for Exact prediction.

Parent-generation record. The ECD parent came from generation batch 2, output 67. That batch began with random seed 301; the individual parent's realized seed is not established by this batch value. Original LigandForge parent coordinates were unavailable for a generator-to-prediction comparison.

Complete selected sequence, from N terminus to C terminus:

```
SSSSSVPAPSIDYLGAFREHGANAEVAEFLYRYLREIKGKSEKEAILLLTAAASIANLVAGISSSVPD
PRTPPPPPPPVPVEPLDPSSLTEVSGATIDLRVKTCVDFKQTVNFFKLRNTQLPGSCTKDNTVLFLLE
GLKLGYASRYFSIGSGSGSQQFS
```

5.20 LK_P093_03 - predicted properties

Boltz2 Exact predicted the full 159-residue candidate against the 440-residue receptor using seed 101. DeltaForge predicted Kd = 0.823 nM for that complete candidate-receptor complex. This is a computational estimate; measured binding and receptor activation remain unknown.

Calculation	Property	Result
Boltz2 Exact	Interface confidence, ipSAE	0.898
Boltz2 Exact	Interface confidence, ipTM	0.886
Boltz2 Exact	Directional ipSAE: candidate to receptor / receptor to candidate	0.898 / 0.588
Boltz2 Exact	Full-fusion / linker pLDDT (0-100)	52.4 / 42.3
Boltz2 Exact	ECD-domain / LigandForge-domain pLDDT (0-100)	68.2 / 43.1
Boltz2 Exact	pDockQ / pDockQ2 / LIS	0.401 / 0.173 / 0.506
Design compared with Exact	Fusion DockQ	0.273
Design compared with Exact	Fusion TM-score	0.4
Design compared with Exact	Receptor-core RMSD (Å)	0.8
Boltz2 Exact	Closest E34 / Q37 / R40 side-chain distances (Å)	3.2 / 2.9 / 2.7
Boltz2 Exact	Free C-terminal height in the membrane reference (Å)	64.0
Boltz2 Exact	Longest peptide C-N bond (Å) / bonds outside 1.1-1.8 Å	1.3 / 0
DeltaForge	Predicted Kd (nM)	0.823
DeltaForge	Contacts / hydrogen bonds / salt bridges / hydrophobic contacts	111 / 9 / 7 / 34
Within this design group	Predicted-affinity rank among the scored top 50	5

Interpretation. The extracellular-position check passed. Segment confidence and agreement with the design reference should be considered alongside the overall interface score: a well-scored interface can coexist with uncertain linkers or domain placement. The model does not establish experimental tag accessibility, affinity or activation.

This fusion exceeds 140 amino acids and was labeled as a longer-sequence case by the affinity provider. No candidate-specific uncertainty estimate was returned.

5.20 LK_P093_03 - bound structures

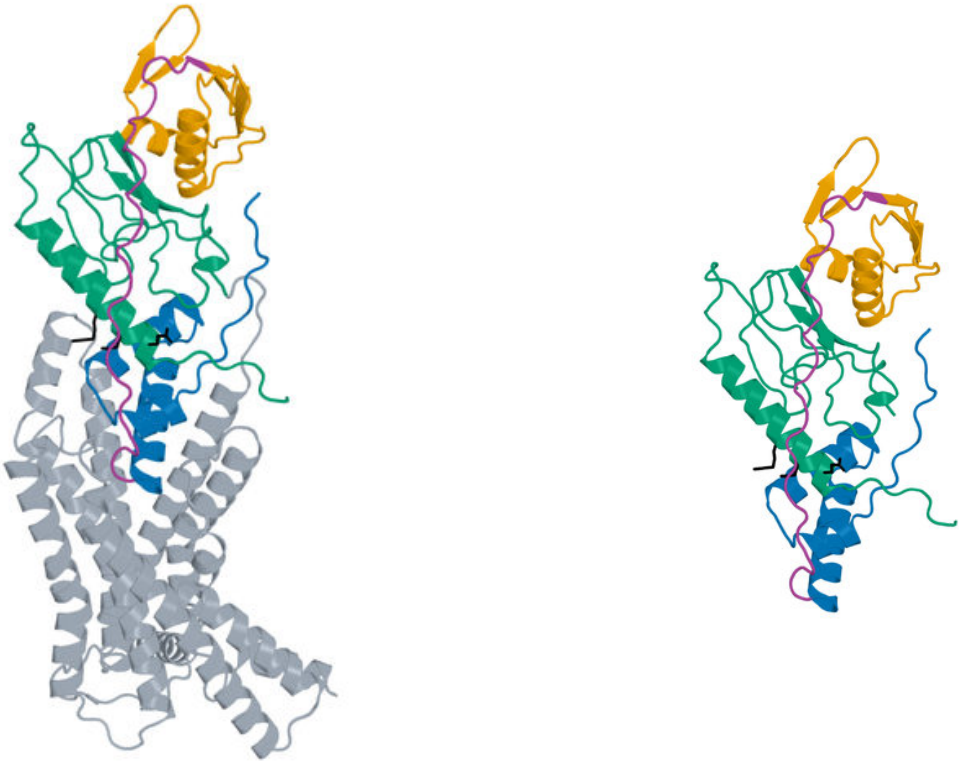


Figure 20. LK_P093_03, final rank 20. Boltz2 Exact prediction using seed 101, with the complete candidate as chain A and the 440-residue receptor as chain B. Left: full complex. Right: interface detail. The LigandForge domain is orange, linker purple and ECD-binding domain blue. Receptor ECD residues B1-105 are teal; the rest of the receptor is gray. Black sticks show E34, Q37 and R40. The panels use different zoom levels. Distant receptor regions are hidden in the detailed display, while the scored model retains all residues.

6 What the predictions support

The final 20 are candidates for synthesis and testing. High interface confidence and low predicted Kd do not establish binding experimentally, and neither demonstrates receptor activation. The different receptor constructs, unresolved original target-pocket mapping, uncertain fusion segments and missing original generated structures limit interpretation.

Parent-domain predictions and fusion predictions answer different questions. Joining two domains can change their structures or receptor placement, so the standalone parent scores should be read alongside the fusion's own confidence and structural-consistency results. No experimental EC50, maximal activation, binding Kd, selectivity, expression or stability result is available for these candidates.

Five further designs passed both selection thresholds but fell below the final-rank cutoff:

Design	Exact ipSAE	Predicted Kd (nM)
LK_P124_07	0.898	0.756
LK_P096_01	0.898	3.62
LK_P111_02	0.897	2.32
Design_352	0.896	0.758
LK_P133_03	0.895	0.728

They were excluded because their ipSAE scores were lower than that of the twentieth selected design. A lower predicted Kd did not override the confidence-first ranking rule.

6.1 Data availability

The report and scientific evidence are available at benavt/adaptyv-week3 on GitHub. All 20 candidates retain the single_chain submission classification and the order shown in section 4.

7 Complete receptor inputs and design history

The sequences below are wild-type human GLP1R, UniProt P43220-1. Candidate sequences are printed in section 5. The original-design predictions paired each candidate with the 409-residue receptor below. ECD-parent and fusion predictions used the 440-residue receptor. For linker generation, the candidate-specific receptor intervals in section 5 specify which fragments of that 440-residue sequence were supplied to AP Novo. Positions are inclusive and count from the first residue shown.

7.1 Original-design receptor: 409 amino acids

```
RPQGATVSLWETVQKWREYRRQCQRSLTEDPPPATDLFCNRTFDEYACWPDGEPGSFVNVSCPWYLPW
ASSVPQGHVYRFCTAEGLWLQKDNSSLPWRDLSECEESKRGERSSPEEQLLFLYIIYTVGYALSFSAL
VIASAILLGFRHLHCTRNYIHLNLFASFILRALSVFIKDAALKWMYSTAAQQHQWDGLLSYQDSLSCR
LVFLLMQYCVAANYYWLLVEGVYLYTLLAFSVLSEQWIFRLYVSIGWGVPLLFVVPWGIVKYLYEDEG
CWTRNSNMNYWLIIRLPILFAIGVNFLIFVRVICIVVSKLKANLMCKTDIKCRLAKSTLTLIPLLGTH
EVIFAFVMDEHARGTLRFIKLFTELSFTSFQGLMVAILYCFVNNEVQLEFRKSWERWRLEHLHIQRDS
S
```

7.2 ECD-parent and fusion receptor: 440 amino acids

```
RPQGATVSLWETVQKWREYRRQCQRSLTEDPPPATDLFCNRTFDEYACWPDGEPGSFVNVSCPWYLPW
ASSVPQGHVYRFCTAEGLWLQKDNSSLPWRDLSECEESKRGERSSPEEQLLFLYIIYTVGYALSFSAL
VIASAILLGFRHLHCTRNYIHLNLFASFILRALSVFIKDAALKWMYSTAAQQHQWDGLLSYQDSLSCR
LVFLLMQYCVAANYYWLLVEGVYLYTLLAFSVLSEQWIFRLYVSIGWGVPLLFVVPWGIVKYLYEDEG
CWTRNSNMNYWLIIRLPILFAIGVNFLIFVRVICIVVSKLKANLMCKTDIKCRLAKSTLTLIPLLGTH
EVIFAFVMDEHARGTLRFIKLFTELSFTSFQGLMVAILYCFVNNEVQLEFRKSWERWRLEHLHIQRDS
SMKPLKCPTSSLSSGATAGSSMYTATCQASCS
```

7.3 How the designs progressed through the workflow

The original population was generated with LigandForge and evaluated with Boltz2 Exact. Its generation settings were documented on 16 October. Exported creation timestamps conflict with an earlier recorded folding completion, so the precise timestamp order remains unresolved.

The ECD-parent campaign was configured on 19 October. Saved AP Novo outputs were recovered during the 21 October processing repair, then screened with Boltz2 Fast and taken forward for the 23 October standalone Exact evaluation. Recovery and transport steps did not create new scientific designs.

On 24 October, the selected ECD and LigandForge parents were paired, and AP Novo generated intervening linkers. Removing the two prohibited pairs left 183 active pairs and 1,830 candidate fusions. Complete candidates were predicted again with the full 440-residue receptor using Boltz2 Exact. The 25 October ranking combined the saved interface scores with DeltaForge results for the two scored groups of 50. The final filter and ranking produced the 20 sequences in this report.

7.4 Remaining provenance gaps

The original LigandForge request and the provider's actual input preparation used different receptor structures and differently numbered pocket lists. Their relationship cannot be reconstructed from the available records. The original submitted receptor structure and original generated complexes are unavailable. The known versions, settings, literal pocket lists and exported sequences are described in section 2; missing details are not inferred.

Some software records identify historical implementations without providing reliable version numbers for every program. This report names the programs, provides the available versions and describes their inputs and settings. It documents the completed calculations without claiming that every historical execution was independently repeated.